
Use Case: Smart Rule-Based Approval Engine

Overview

The Smart Rule-Based Approval Engine is designed to automate and streamline common organizational approval workflows such as expense claims, leave requests, and discount approvals. The system evaluates requests based on predefined rules linked to an employee’s role (grade) and determines whether the request can be auto-approved or requires manual intervention.

Approval Logic

Each employee grade has a predefined annual pre-approval limit for different categories. Requests that fall within these limits are automatically approved. If a request exceeds the allocated limit, it is escalated to the reporting manager for review.

Important Notes:

- All pre-approved limits are annual, not monthly.
 - Employees can choose how to utilize their annual limits (e.g., monthly claims or a single bulk request).
 - Any request exceeding the annual threshold must go through the approval hierarchy.
-

Sample Guidelines (Shared with Interns)

Grade	Expense Limit (Annual)	Leave Limit (Annual)	Discount Limit
-------	---------------------------	-------------------------	-------------------

Grade 1	₹10,000	30 days	20%
Grade 2	₹25,000	36 days	30%
Grade 3	₹50,000	45 days	40%

Role Hierarchy (Sample Organization Structure)

- **Admin**
 - └─ Sophia Carter
 - **Manager**
 - └─ Lee Johnson
 - **Employee**
 - └─ Liam Wilson (Grade 1)
-

Acceptance Criteria

- The system supports three employee grades: Grade 1, Grade 2, and Grade 3.
 - Request categories (Expense, Leave, Discount) are predefined and configurable.
 - Requests within pre-approved limits are automatically approved.
 - Requests exceeding pre-approved limits are routed to the Manager.
 - The Admin has the authority to:
 - Approve or reject Manager-level requests.
 - Review any request that exceeds defined approval rules.
-

Functional Requirements

The system must support the following request types:

- **Leave requests**
 - **Expense claims**
 - **Discount requests**
-

System Behavior

- **Users can submit requests via the web application.**
 - **The system evaluates each request using predefined approval rules.**
 - **Requests are processed in one of the following ways:**
 - **Automatically approved**
 - **Automatically rejected**
 - **Assigned to an approver for manual review**
 - **Approvers can review assigned requests and take appropriate action.**
 - **After a decision is made, the system updates the request status accordingly.**
 - **Each request clearly displays the reason for approval or rejection.**
-

Admin Capabilities

- **Create, modify, and manage approval rules.**
- **View all requests and their approval status.**
- **Access reports and analytics related to:**
 - **Rule usage**
 - **Approval patterns**
 - **Request statistics**